

The Processors

Cortex A series

The Cortex A series is the code name for the processor reference design from ARM and the series includes Cortex A5, Cortex A7, Cortex A8, Cortex A9, Cortex A12 and Cortex A15 reference designs with the A5 at the entry level side and the A15 meant for the high-performance mobile devices. Apart from being found in smartphones and tablets, the Cortex A series processors are also seen in digital TVs, media players, wireless networking devices, servers and so on. But for the purpose of the following pages we will only concentrate on their use in smartphones and tablets.

Cortex A15 is the flagship processor in the Cortex A series lineup and is meant for high-performance higher-end mobile devices. It is designed with advanced power reduction techniques and features an out-of-order superscalar pipeline for faster execution of instructions. The Cortex A15 will be bundled with other low power cores, to keep in line with ARM's big. LITTLE technology. This is similar to NVIDIA Tegra's 5th companion core. It basically allows the A15 to take care of the higher end tasks such as gaming, number crunching, rendering and so on while a lower powered processor core will take care of simpler tasks such as web surfing, background tasks in idle times and so on.

Krait

Qualcomm is the only company among the SoC makers which came out with its own processor for its SoCs. Krait was the successor to the hugely popular Scorpion processor and was first seen in the Snapdragon S4 lineup of Qualcomm SoCs. While the Scorpion could only be used in single and dual-core configurations, the Krait can be used in single, dual and quad-core configurations. Based on the 28nm manufacturing process, the Krait CPU was refreshed in 2013 when Qualcomm announced the Krait 300 and Krait 400 which will be seen in future Snapdragon devices. The Krait 300 will run at 1.9GHz whereas the Krait 400 can run at upto 2.3GHz. The Krait 300 will cater to more mainstream devices and will be seen in the Snapdragon 400 and Snapdragon 600 SoC sporting devices, whereas the higher end Krait 400 will be seen in the Snapdragon 800 SoC. 